

E-Commerce Conversion Analytics

Portfolio Case Study · Jibin Kurian

Business question	How effectively does the e-commerce platform turn customer browsing activity into purchases, and what factors are associated with conversion?
Analysis type	Observational analysis of recorded e-commerce events from May–September 2015. The analysis identifies patterns and associations; it does not establish causation.

1. Executive Summary

This project analyses recorded e-commerce behaviour to understand how visitors move from product browsing toward cart activity and purchase. It covers the conversion funnel, customer activity, daily behaviour, category performance, product performance, item availability, and the quality of historical attribute matching.

The dataset contains 2,756,101 recorded events from 1,407,580 visitors. The largest observed funnel drop occurs between viewing a product and recording an add-to-cart event: the stage-based View → Cart rate is 2.69%. The stage-based Cart → Purchase rate is 31.07%, and the overall stage-based View → Purchase rate is 0.83%.

A separate simplified sequential analysis found 9,682 visitors completing View → Cart → Purchase, equivalent to 0.69% of all visitors. Availability is substantially associated with observed conversion: 1.71% for Available, 0.16% for Unavailable and 0.40% for Unknown / Not Matched. These differences are observational and should not be read as causal effects.

The main areas for investigation are the View → Cart drop-off, availability-related browsing behaviour, high-traffic and low-conversion products and categories, event-tracking consistency, and incomplete historical attribute coverage.

2. Business Problem

An e-commerce platform records large volumes of customer interaction data, but raw event volume alone does not explain whether browsing is turning into purchases. The business needs to understand where visitors leave the observed funnel, which areas deserve deeper investigation, and whether item attributes such as availability are associated with different observed conversion levels. The project asks two connected questions:

- How effectively does recorded browsing activity translate into purchase activity?
- Which observable factors are associated with differences in conversion?

3. Dataset & Scope

The analysis uses the Retailrocket Recommender System Dataset, repurposing its recorded interaction and item-property data for an e-commerce conversion analytics use case.

Dataset component	Role in analysis
events.csv	Visitor interaction events: view, addtocart and transaction
category_tree.csv	Category hierarchy and parent-child relationships
item_properties_part1.csv	Historical item properties, including category and availability
item_properties_part2.csv	Second historical item-property slice

The recorded event period runs from 3 May 2015 to 18 September 2015 (139 days). The events table contains 2,756,101 rows, 1,407,580 unique visitors and 235,061 unique items.

4. Data Investigation & Quality Challenges

The raw data was not treated as a clean transactional table. Several structural issues had to be understood before analysis.

- **Event composition:** 2,664,312 view events, 69,332 add-to-cart events and 22,457 transaction events.
- **Transaction ID missingness:** transactionid is missing for normal view and add-to-cart events by design; it is not automatically a data error.
- **Duplicate events:** exact duplicate event records were identified and investigated rather than blindly deleted. The enriched events table preserved all 2,756,101 original event rows.
- **Transaction events vs orders:** 22,457 transaction event rows correspond to 17,672 unique transaction IDs, so the transaction-event count is not the same as an order count.
- **Historical item properties:** item properties contain 18 timestamps, so category and availability were treated as time-varying attributes, not static fields.
- **Category enrichment:** category matching was possible for 76.16% of events.
- **Availability enrichment:** availability matching was possible for 80.71% of events.
- **Tracking sequence anomalies:** some purchases have no preceding recorded cart and some cart users have no recorded purchase, so event sequences must be interpreted carefully.

5. Data Preparation & Time-Aware Enrichment

The item-property files were investigated separately and combined after checking their overlap. The two files contained the same set of timestamps but no overlapping unique item-property-timestamp records, so they could be concatenated without treating the second file as a duplicate copy.

The key methodological requirement was to avoid attaching a future property value to an earlier event. Historical property records were therefore matched using event time, with separate time-aware enrichment for category and availability. The enriched events table preserved all 2,756,101 original rows, and no future category or availability matches were found. Events without a safe historical match were kept with NULL enrichment values, not labelled as missing attributes or unavailable items.

6. Data Model & MySQL

The cleaned and enriched data was loaded into MySQL for structured SQL analysis, using a simple fact/dimension model:

- **fact_events:** the central event table, holding event-level activity and the time-aware category and availability attributes.
- **dim_category:** the category hierarchy (categoryid, parentid).

A foreign key connects fact_events.categoryid to dim_category.categoryid, and dim_category.parentid references dim_category.categoryid to preserve the hierarchy. This separates high-volume event data from reusable hierarchy information.

7. Analytical Framework

The SQL analysis was organised into eight business themes and 23 queries (listed in Appendix A).

Theme	Business purpose	Queries
1. Overall funnel	Visitors reaching view, cart and purchase stages	Q1-Q3
2. Sequential funnel	The observed chronological View → Cart → Purchase path	Q4
3. Customer behaviour	Visitor activity depth and high-activity behaviour	Q5-Q9
4. Time analysis	Daily traffic, event mix and daily observed conversion	Q10-Q12
5. Category analysis	High-activity and high/low observed conversion categories	Q13-Q15
6. Availability analysis	Observed conversion across availability states	Q16-Q17
7. Product analysis	Highly viewed, high-conversion and investigation-candidate products	Q18-Q21
8. Data quality / tracking	Enrichment coverage and sequence limitations	Q22-Q23

8. Key Findings

8.1 Funnel performance (stage-based)

Stage-based counts show how many visitors recorded each type of event. A visitor can record a purchase without a recorded cart event, so these counts are not a strictly sequential journey.

- 1,404,179 visitors recorded a view event; 37,722 recorded an add-to-cart event; 11,719 recorded a transaction event.
- View → Cart: 2.69% (cart users ÷ viewers). Drop-off: 97.31%.
- Stage-based Cart → Purchase: 31.07% (purchasers ÷ cart users). Drop-off: 68.93%.
- Stage-based View → Purchase: 0.83% (purchasers ÷ viewers).

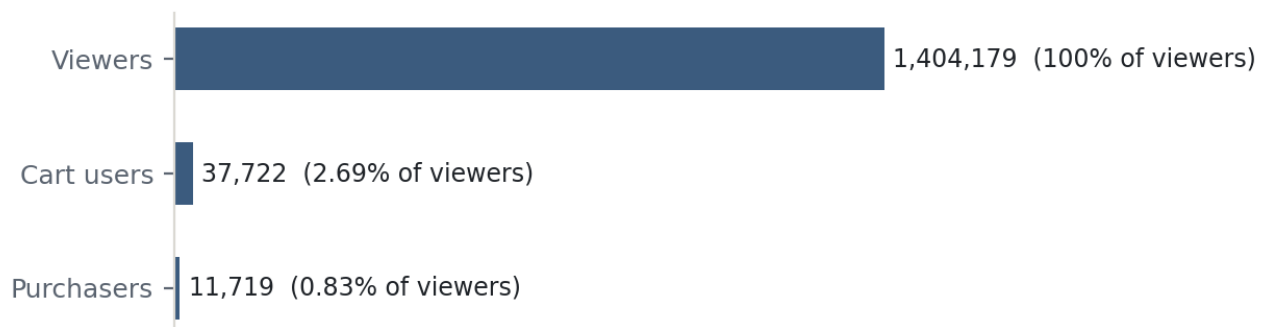


Figure 1. Observed funnel stage reach: visitors with at least one recorded event of each type (Q1, Q2). Stage-based, not a strictly sequential journey.

The largest observed drop is at View → Cart, which makes the move from product interest to cart activity the primary investigation area.

Reading the 31.07% carefully. The stage-based Cart → Purchase rate is purchasers ÷ cart users. Because 1,272 purchases have no preceding recorded cart (Q6), it is not the same as the share of cart users who purchased. Of 37,722 cart users, 27,275 have no recorded purchase (Q5), so 10,447 do (27.69%; derived as Q1 cart users minus Q5). The two measures answer different questions and are reported separately.

8.2 Sequential journey

- 9,682 visitors followed the simplified observed sequence View → Cart → Purchase, 0.69% of all visitors. Sequential analysis uses simplified event ordering.
- 27,275 cart users had no recorded purchase (Q5). This may reflect abandonment, incomplete tracking, or other event-ordering limitations.
- 1,272 purchases had no preceding recorded cart (Q6). This may indicate missing or incomplete cart-event tracking, or a different recorded event sequence.

8.3 Customer behaviour and daily activity

- The dataset contains an average of 1.96 events per visitor; 71.15% of visitors recorded only one event and 28.85% recorded several.
- Most visitors therefore have very shallow recorded histories. High-activity visitors are useful for investigation, but high activity alone does not prove customer value, bot behaviour or fraud.
- Observed daily view-to-purchase conversion ranged from 0.33% to 1.31% across the 139 days (Q12).



Figure 2. Observed daily view-to-purchase conversion, purchasers ÷ viewers per day (Q12).

8.4 Availability

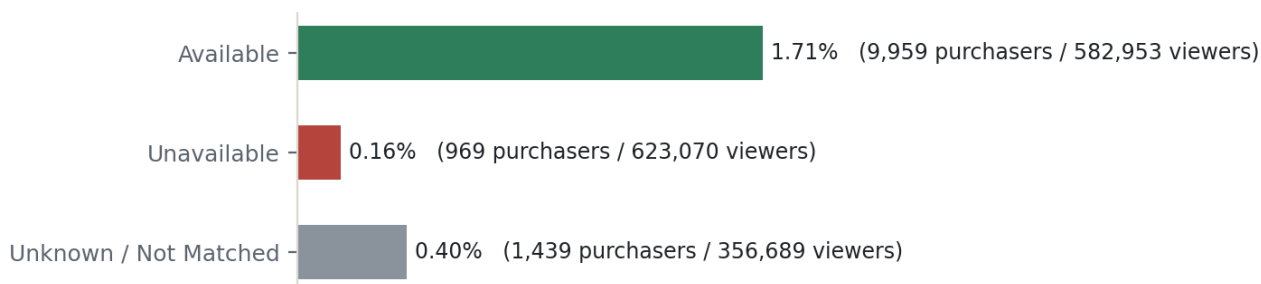


Figure 3. Observed view-to-purchase conversion by availability status (Q17). Observational; viewers and purchasers overlap across statuses and are not additive.

Available items show 1.71% observed conversion, Unavailable 0.16% and Unknown / Not Matched 0.40%. The difference is substantial and makes availability an important investigation variable. It is observational: the analysis does not show that availability caused the difference. Unknown / Not Matched means no safe historical availability match was found, not that the item was unavailable. Visitors can appear in more than one availability status, so no significance test is reported.

8.5 Category and product investigation

The category and product analyses identify areas for investigation, not winners and losers. Q14 lists categories with higher observed view-to-purchase rates, Q15 lists categories with many viewers and no recorded purchase, and Q18–Q21 perform comparable investigations for products. Category and item IDs are used because reliable names are not available in the source data.

For example, item 416017 recorded 186 viewers and 32 purchasers, an observed view-to-purchase rate of 17.20% (Q19). This is an observed rate within the dataset, not proof that the product is inherently better or that it would generalise beyond the period.

Conversion lists are small-sample sensitive, so each shows the sample beside every rate. The smallest samples in the outputs are 52 viewers in Q14, 103 in Q15 and 52 in Q19, 571 unique viewers in Q20 and 20 cart users in Q21.

8.6 Data quality and coverage

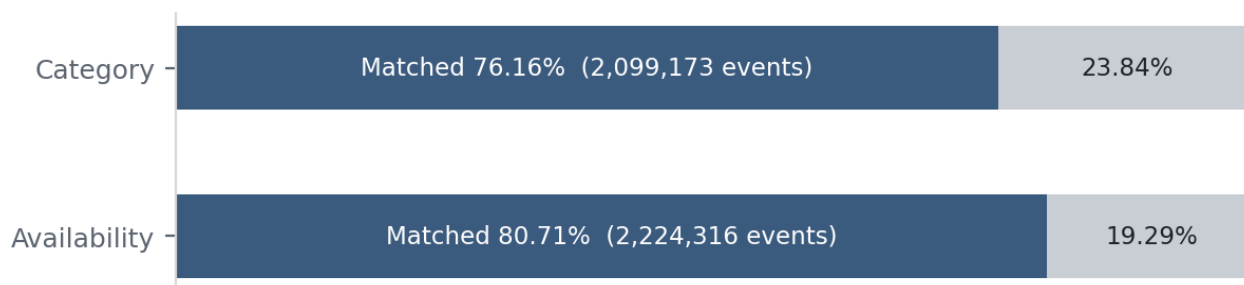


Figure 4. Share of events with a safe historical match (Q22, Q23). Not matched means no safe historical property match was found; it does not mean missing, unavailable or poor.

9. Key Analytical Decisions

- **Visitors, not events, for the funnel.** Stage-based rates use visitor counts, and transaction events are not treated as orders (22,457 events vs 17,672 transaction IDs).
- **Two funnels, kept separate.** The stage-based funnel (Q2) and the simplified sequential funnel (Q4) answer different questions and are never combined into one rate.
- **Time-aware enrichment.** Category and availability were attached using event time so that no future value was applied to an earlier event.
- **Unmatched stays NULL.** Events without a safe historical match were kept with NULL values, not recoded as unavailable or missing.
- **Observational language.** Findings are described as observed rates and associations, with sample sizes shown beside rankings.

10. Dashboard

The dashboard presents the analysis as a stakeholder-facing analytical product, not a collection of SQL outputs. It includes an executive overview, conversion funnel, customer behaviour, category performance, product performance, and availability and data-quality pages. It is driven by the Q1-Q23 outputs: each analytical card shows its source query, and the data layer is generated from the CSV outputs rather than typed into the interface. The visual style is deliberately restrained: neutral background, white cards, subtle borders and limited colour coding.

11. Business Recommendations

Each recommendation is an investigation priority, with a suggested measure of progress. None is a forecast of revenue impact.

Recommendation	What to look at	How progress could be tracked
Investigate the View → Cart drop-off	Product-page behaviour, category differences and availability status	View → Cart rate by category and by availability status
Investigate availability-related browsing	How much activity lands on unavailable items, and whether unavailable inventory is concentrated in particular categories or products	Share of events on Unavailable items (Q16) and its conversion (Q17)
Review high-traffic, low-conversion candidates	Q15 and Q20 as investigation lists, not conclusions about poor performance	Conversion of listed items after any change, with sample size
Investigate event-tracking consistency	The 27,275 cart users without a recorded purchase and the 1,272 purchases without a preceding recorded cart	Counts in Q5 and Q6 after tracking checks
Improve historical attribute coverage	Safe category and availability matching, to reduce the Unknown / Not Matched population	Matched share of events (Q22, Q23)

12. Limitations

- This is observational event data; the analysis does not establish causality.
- The stage-based and simplified sequential funnels answer different questions and are not interchangeable.
- Category matching covers 76.16% of events and availability matching 80.71%, so some events cannot be safely segmented by these attributes.
- Visitors can appear in more than one availability status, so no significance test is reported for the availability comparison and it is not controlled for category.
- The source provides no reliable human-readable category or product names.
- There is no revenue or profit information, so financial impact cannot be quantified.
- Event-tracking limitations are visible through carts without recorded purchases and purchases without preceding recorded carts.
- The dataset represents recorded behaviour; unrecorded actions cannot be inferred.

13. Final Conclusion

The largest observed conversion challenge is the transition from product viewing to cart activity. Only 2.69% of viewers recorded an add-to-cart event. Purchasers equal 31.07% of cart users under the stage-based definition (27.69% of cart users have a recorded purchase when counted strictly), giving 0.83% observed View → Purchase conversion. The simplified sequential funnel completed for 0.69% of all visitors.

Availability is substantially associated with observed conversion, although the analysis cannot establish that availability caused the difference. Category and product analysis provide further investigation candidates, while incomplete historical enrichment and event-sequence anomalies show that tracking quality must be considered alongside business behaviour.

The project turns a large raw event dataset into a structured workflow: data investigation → time-aware enrichment → MySQL data model → SQL business analysis → dashboard → business recommendations.

14. Project Stack

Data	Retailrocket e-commerce event and item-property data
Python	Data profiling, investigation, cleaning and time-aware enrichment
MySQL	Relational data model and SQL analysis
SQL	23 business-focused analytical queries
Dashboard	HTML, CSS and JavaScript, generated from the query outputs

Appendix A. Query Index

Every figure in this case study and in the dashboard is traceable to one of these SQL outputs.

Query	Business question answered
Q1	Overall visitor funnel counts
Q2	Funnel conversion rates
Q3	Funnel drop-off
Q4	Sequential View → Cart → Purchase
Q5	Cart users without recorded purchase
Q6	Purchases without preceding recorded cart
Q7	Average events per visitor
Q8	One-event vs multiple-event visitors
Q9	Most active visitors
Q10	Daily activity
Q11	Daily event types
Q12	Daily visitor-to-purchase conversion
Q13	Category activity
Q14	Category observed conversion
Q15	Categories with many views and few purchases
Q16	Availability activity
Q17	Availability vs observed conversion
Q18	Products with most views
Q19	Products with strongest observed view-to-purchase conversion
Q20	Products with many views and no recorded purchase
Q21	Products with strong cart activity but relatively lower cart-to-purchase conversion
Q22	Category enrichment completeness
Q23	Availability enrichment completeness